KETCo Unified Digital Platform

REQUIREMENTS CATALOGUE

ID	Subsystem	Name	Description	Priority	Rationale
FR-101	Intake	Unique ID Generation	The system must generate a unique reference ID for each sample at receipt.	Must	Rationale: Fundamental operational dependency. Verify: Inspection of data rows.
FR-102	Intake	Spatial Mapping	The system must record GPS coordinates and link them to a parent location ID.	Must	Rationale: Solves geographical naming ambiguity. Verify: Entry mapping demo.
FR-103	Intake	Condition Status	Let technician set status to 'Provisionally Accepted' and force note entry.	Must	Rationale: Handle incomplete field data safely. Verify: Null note rejection test.
FR-104	Intake	Suitability	Check temperature, container, and elapsed time against rules.	Should	Rationale: Prevent testing degraded samples. Verify: Test edge metrics.
FR-105	Tracking	Splits	Support sample splits and link to parent reference key.	Must	Rationale: Mandatory for external referrals. Verify: Inspect DB relations.
FR-106	Tracking	Immutable Log	Add log row automatically when location or staff ID changes.	Must	Rationale: Eliminate mix-ups and log failures. Verify: Delete/update blocking.
FR-107	Lab Ops	Segregation	Separate raw data from reportable fields until review is done.	Must	Rationale: Prevent customer viewing unverified values. Verify: Portal isolation test.
FR-108	Lab Ops	Validation	Verify analyst ID against certified methods list before save.	Should	Rationale: Quality assurance check. Verify: Uncertified save block.
FR-109	Lab Ops	Machine Block	Set test to 'Delayed' if machine fails QC check or is out of service.	Must	Rationale: Prevent execution on uncalibrated hardware. Verify: Status shift demo.
FR-110	Reports	Version Ctrl	Increment version number and force revision notes during corrections.	Must	Rationale: Vital for audit trails and liability protection. Verify: Blank note block test.
FR-111	Reports	Dashboard	Show unapproved corrected reports, anomalies, referrals, and delays.	Should	Rationale: Live operational visibility for managers. Verify: UI rendering check.
NFR-201	Offline	Sync Cap	Store 200 logs offline and sync in under 30 seconds.	Must	Rationale: Counteract weak rural networks. Verify: Cache sync simulation.
NFR-202	Security	Multi-Tenant	Filter queries by sensitivity flag; return HTTP 403 error.	Must	Rationale: Absolute multi tenant isolation. Verify: Token injection attempt.
NFR-203	Alerts	Latency	Send email/SMS within 5 minutes when Rejected or Delayed.	Should	Rationale: Improve response speed. Verify: Timestamp log check.
NFR-204	Scope	Isolation	Zero software dependencies on billing or enforcement modules.	Must	Rationale: Enforce defined brief boundaries. Verify: Architecture tree inspection.

Strict Operational Boundaries (Out-of-Scope Constraints)

The system must maintain absolute code isolation from non-core processes. The following parameters are formally outside the application architecture boundaries:

1. **Machine Connections:** No direct hardware APIs or software drivers. Analysts must transfer data manually.
2. **Regulatory Compliance:** No automated evaluation of statutory water/soil thresholds or immediate notifications to government agencies.
3. **Financial Operations:** Zero infrastructure supporting invoices, customer pricing records, or active billing portals.